

## Unit 13, Lesson 7: Finding Missing Angles in Polygons

### Benchmark

MA.8.GR.1.6 Develop and use formulas for the sums of the interior angles of regular polygons by decomposing them into triangles.

### Learning Target

- ☐ I can write and solve algebraic equations to find the measure of an interior angle of a  $n$ -gon.

### 13.7.1 Warm-Up: Fighting Fire with Math

**Cassie Pennicuk fights fire with math. As a program manager at the County Fire Authority, her job is to lead the growth program. Part of her job is to create a bridge between the technical, strategic, and tactical teams in the organization. She says, "If you really want something in life, go for it and celebrate the effort. In the end, hard work definitely pays off."**

1. What do you think she means by "celebrate the effort"?
2. Describe a time you went for something that didn't work out, but you still celebrated the effort.

13.7.2 Guided Instruction: Measures of Interior Angles Using Equations

1. The measure of an angle in a regular 24-sided polygon is  $(12x + 6)^\circ$ .

Find the value of each.

| Description                   | Value |
|-------------------------------|-------|
| Sum of the interior angles    |       |
| Measure of one interior angle |       |
| $x$                           |       |

2. A regular polygon can be decomposed into 43 triangles. The measure of one angle of the polygon is  $(17x + 2)^\circ$ . Find the value of  $x$ .
3. The measures of two interior angles of a regular polygon are  $(9x + 48)^\circ$  and  $(15x - 24)^\circ$ , respectively.
- a. Discuss with your partner how to use the given information to find the value of  $x$ .
  - b. Write the equation that can be used to find the value of  $x$ .
  - c. Solve the equation.

d. What is the angle measure for each interior angle in the polygon?

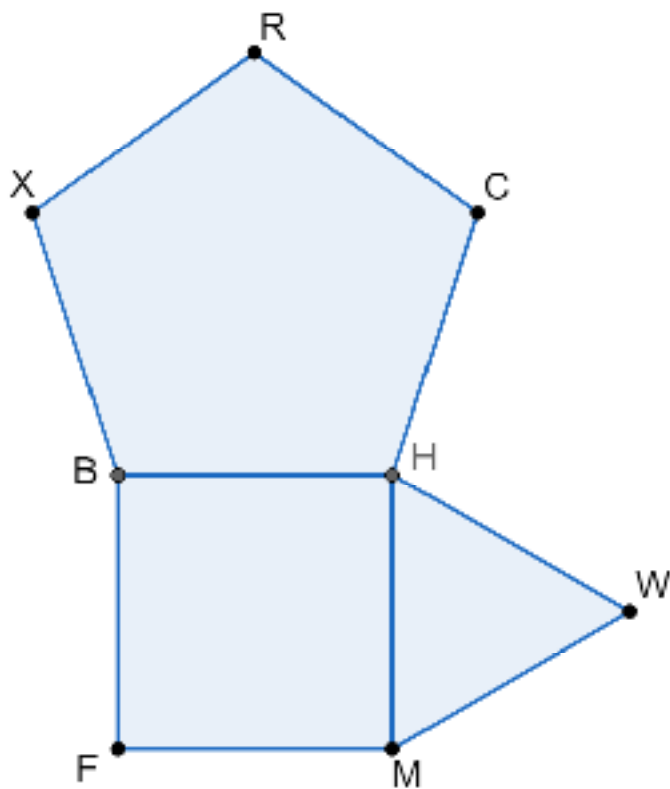
4. Discuss with your partner how to determine the number of sides that a regular polygon has if the sum of the interior angles is given.

a. Work with your partner to complete the table.

| Sum of the Interior Angles | Number of Sides |
|----------------------------|-----------------|
| $360^\circ$                |                 |
| $540^\circ$                |                 |
| $720^\circ$                |                 |
| $900^\circ$                |                 |

b. How many sides does a regular polygon have if the sum of the interior angles is  $6,300^\circ$ ?

5. Regular polygons  $XRCHB$ ,  $BHMF$ , and  $HWM$  are shown.



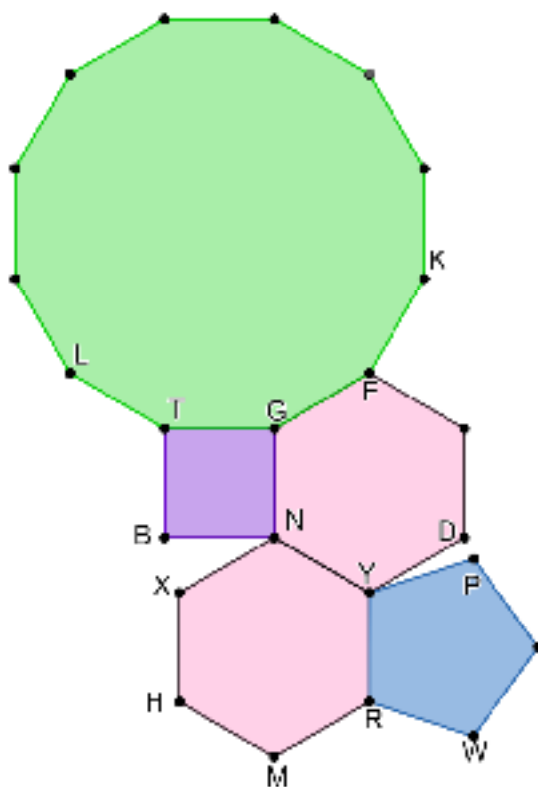
The measure of angle  $WHC$  is  $(19x - 27)^\circ$ . Determine the value of  $x$ .

### 13.7.3 Guided Practice: Measures of Interior Angles Using Equations

1. How many sides does a regular polygon have if the sum of the interior angles is  $10,620^\circ$ ?
2. The measures of two interior angles of a regular polygon are  $(6x + 18)^\circ$  and  $(10x - 86)^\circ$ , respectively. What is the angle measure for each interior angle?
3. A regular polygon can be decomposed into 23 triangles. The measure of one angle of the polygon is  $4.5x + 36$ . Find the value of  $x$ .

**13.7.4 Your Turn!**

1. The measure of an angle in a regular 50-sided polygon is  $(12.8x - 22.4)^\circ$ . Find the value of  $x$ .
  
2. Five regular polygons are shown.



- a. The measure of angle  $MRW$  is  $(16x - 44)^\circ$ . Determine the value of  $x$ .
  
- b. The sum of the angles  $LTG$ ,  $TBN$ ,  $NYD$ , and  $WRY$  is  $(8m + 52)^\circ$ . Find the value of  $m$ .

**13.7.5 Wrap-Up: Reflection**

1. **Reflect on your learning today by completing each sentence. Write your response on each finger.**

